

Vira Rithy



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Portfolio



@rithyvira



Phnom Penh, Cambodia

PROFILE

Passionate about Data Science, AI, GenAI, and Machine Learning, with expertise in machine learning, data analytics, and cutting-edge AI applications. Skilled in extracting insights from complex datasets and delivering impactful solutions. Driven by innovation, collaboration.

EDUCATION

DataU Academy

Data Specialist

08/2023 - 01/2024

Institute of Technology of Cambodia

Bachelor of Engineering (BEng), Data Science

02/2021 - 07/2025

CERTIFIED LICENSE

SDOE : DevOps Engineering

SalaCyber

SHOW CREDENTIAL

IBM AI Engineering Specialization

IBM

SHOW CREDENTIAL

AI Engineering Bootcamp

Maven - AI Makerspace

SHOW CREDENTIAL

Machine Learning with Python-From Linear Models to Deep Learning

MITx MicroMasters

SHOW CREDENTIAL

PERSONAL PROJECTS

Multi-Agent RAG Marketing Content Enhancer"

The Multi-Agent RAG platform drafts, edits, cites, and adapts, ensuring accurate content with future visual and scalable features.

OpenAI

LANGCHAIN

LANGGRAPH

SHOW PROJECT



PROFESSIONAL EXPERIENCE

ZTOA Group

AI Engineer

04/2024 - Present

- Email Application System Dashboard** NODE.JS EXPRESS Javascript POSTGRES SHOW PROJECT
Built a full-stack AI-powered email dashboard that imports contact emails, generates personalized content from domain websites, and automates campaign delivery. Uses fallback templates for generic emails.Tech stack: React, Node.js, Express, PostgreSQL, Apache
- Automated Scrape Email and Generated SEO Marketing Report"** n8n OpenAI
Built an end-to-end n8n workflow to scrape business websites from Google Maps, extract emails, generate structured SEO reports, and auto-send personalized result, using JavaScript, PostgreSQL, OpenAI, and SMTP.

DataTicon

AI Engineer

02/2024 - 03/2025

- Khmer Speech Recognition "Speech-to-Text EN-KH"** PYTORCH HUGGING FACE SHOW PROJECT
Fine-tuned OpenAI Whisper-Small for KH and EN, achieving a 37% WER. Deployed on Hugging Face with a Gradio-integrated UI.
- AI Data Analysis "Text-to-SQL, Voice-to-Chart"** LLM CHAINLIT PYTHON SHOW PROJECT
Built an AI tool using dynamic databases for text-to-SQL and text-to-chart outputs, enabling quick insights and efficient data visualization
- Machine Learning "Forecasting Cambodia Tourism"** ENSEMBLE PYTHON MLFLOW SHOW PROJECT
Achieved 97% accuracy in forecasting monthly visitors (e.g., August 2024: Predicted 548K vs. Actual 547K) using an ARIMA-XGBoost ensemble with MLflow tracking.

DataU Academy

Data Specialist

08/2023 - 01/2024

- Data Warehousing: "Global Cyclistic Company"** SQL ETL SHOW PROJECT
Transform massive datasets from diverse sources for accessible insights and powerful sales optimization analytics.
- AI-Powered Sales Analysis & Forecasting** PYTHON STREAMLIT SHOW PROJECT
Developed a sales forecasting solution for Olist using SARIMAX, XGBoost, and LSTM (top performer), integrating Streamlit and Power BI for revenue, stock, and marketing insights.
- Hazard Vulnerability Dashboard** POWER BI SHOW PROJECT
Analyzed Cambodia's hazard vulnerability index (2014-2019) at provincial/commune levels, assessing risks from storms, floods, and droughts, with Khmer and English support.



SKILLS

TECHNICAL

Programming

Python for Data processing, ML model development, API integration, and PyTorch for deep learning.

DevOps

Experienced in CI/CD, Docker, Kubernetes, and ML pipeline automation using

Power BI

Data visualization, reporting, and ETL tools for data integration & analysis.

Math

Statistics, Probability, Linear Algebra, Probabilistic Graphical Models, Calculus, Numerical Analysis, Discrete Maths

LLMs

Hands-on experience with Prompt Engineering, RAG, Fine-Tuning and Multi-Agent Systems.

INTERPERSONAL

LEADERSHIP

Inspiring others through positive example, effective communication, and collaborative decision-making.

COMMUNICATION

Inspiring others through positive example, effective communication, and collaborative decision-making.

PROBLEM SOLVING

Prioritize analytical thinking, creativity, and a collaborative mindset to find innovative and effective solutions.

BUSINESS CONCEPT

Centers on delivering innovative products/services that address market needs, backed by a commitment to quality, sustainability, and exceptional customer experiences.